Tecniche integrali

1) Lo vedo a occhio (Lo so a memoria / Ho un bigliettino)
2) Aggiungo e tolgo cose al numeratore
3) Sostituzione
4) Se al denominatore ho num + x^2 (e al numeratore un num, lo riporto a $1+x^2$
5) Formule Parametriche

$$t = tg\frac{x}{2} \qquad \sin x = \frac{2t}{1+t^2} \qquad \cos x = \frac{1-t^2}{1+t^2} \qquad dx = \frac{2}{1+t^2}\,dt$$

6) Integrazione per parti

Settimana: 20

Materia: Matematica
Classe: 5A
Data: 23/02/26

Integrali:

Esempio: $\int \frac{1}{1+x}\,dx = \ln(|1+x|) + c$

$\int \frac{1}{1+\sqrt{x}}\,dx$ (Sost.: $\sqrt{x} = t$)

Orango – Tango
▷ Si ricava la x in fz della t

$x = t^2$

▷ Si deriva a sinistra in x, a destra in t e si aggiunge rispettiv. dx e dt

$1 \cdot dx = 2t\,dt$

▷ Ho ricavato dx, lo sostituisco

$$\int \frac{1}{1+t}\,dx = \int \frac{1}{1+t} \cdot 2t\,dt =$$

Linearità.

$$\int \frac{2t}{1+t}\,dt \boxed{=} 2\int \frac{t}{1+t}\,dt =$$

$$2\int \frac{t+1-1}{t+1}\,dt$$

$$\frac{t+1}{t+1} - \frac{1}{t+1}$$

$$2\int \underbrace{\frac{t+1}{t+1}}_{1} - \frac{1}{t+1}\,dt$$

$$2\int \left(1 - \frac{1}{t+1}\right) dt = 2\left[\int 1\,dt - \int \frac{1}{1+t}\,dt\right] =$$

$$2\left[t - \ln|1+t|\right] + c = 2\left(\sqrt{x} - \ln|1+\sqrt{x}|\right) + c$$

Fatto: L'integrale è Lineare, cioè

▷ $\int k f(x)\,dx = k\int f(x)\,dx \qquad k \in \mathbb{R}$

$$\int (f(x) \pm g(x))\,dx = \int f(x)\,dx \pm \int g(x)\,dx$$

Dim: Si comporta come la derivata

Fatto: È lecito fare sostituzioni come per i limiti. Ricordarsi che il dx (differenziale) va modificato in modo opportuno seguendo il metodo Orango-tango

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Tip: Sostituire TUTTA la roba che non vi piace

$$\int \frac{x}{\sqrt{x-1}}\,dx =$$

Sost: $\sqrt{x-1} = t$

$x-1 = t^2 \rightsquigarrow dx = 2t\,dt$

$x = t^2+1$

$$= \int \frac{t^2+1}{\cancel{t}}\,2\cancel{t}\,dt = 2\int (t^2+1)\,dt = 2\left[\frac{t^3}{3} + t\right] + c$$

$$\frac{2}{3}t^3 + 2t + c = \frac{2}{3}\sqrt{(x-1)^3} + 2\sqrt{x-1} + c$$

355: $$\int \frac{1}{x-\sqrt{x}}\,dx =$$

Sost: $\sqrt{x} = t \rightsquigarrow dx = 2t\,dt$

$x = t^2$

$$= \int \frac{1}{t^2-t}\cdot 2t\,dt = \int \frac{1}{\cancel{t}(t-1)}\,2\cancel{t}\,dt = 2\int \frac{1}{t-1}\,dt$$

$$= 2\ln|t-1| + c = 2\ln|\sqrt{x}-1| + c$$

356: $$\int \frac{1}{e^x+e^{-x}}\,dx =$$

Sost $e^x = t \rightsquigarrow dx = \frac{1}{t}\,dt$

$x = \ln(t)$

$$= \int \frac{1}{t+\frac{1}{t}}\cdot\frac{1}{t}\,dt = \int \frac{\cancel{t}}{t^2+1}\cdot\frac{1}{\cancel{t}}\,dt = \operatorname{arctg}(e^x) + c$$

Alternat.

$$\int \frac{1}{e^x+\frac{1}{e^x}}\,dx = \int \frac{e^x}{e^{2x}+1}\,dx = \text{Sost. } e^x = t$$

$$e^x\,dx = dt$$

$$\int \frac{1}{t^2+1}\,dt = \dots = \operatorname{arctg}(e^x)+c$$

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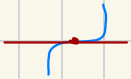
$f(x) = \dfrac{ax^2-1}{x+2}$ trova a in modo che $x=1$ max

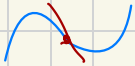
Si impone $f'(1)=0$

$$f'(x) = \frac{(2ax)(x+2)-(ax^2-1)(1)}{(x+2)^2} \qquad f'(1)=0$$

$$0 = \frac{2a\cdot 3-(a-1)}{3^2} \rightsquigarrow 6a-a+1=0$$

$$\rightsquigarrow 5a=-1 \rightsquigarrow \boxed{a=-\frac{1}{5}}$$



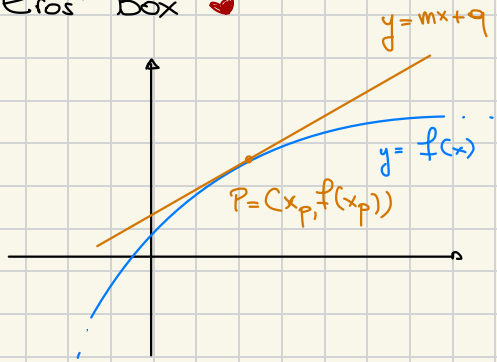


35c $f(x) = e^{\frac{ax-b}{x+c}}$ nel pto di ascissa 0

(1) Ha un flesso $\rightsquigarrow f''(0)=0$

(2) La tangente vale $y=\frac{2}{e}x+\frac{1}{e}$

Eros' Box ♥



(1) $m = f'(x_p)$

(2) $P \in$ Retta, dunque vale:

$y_p = m x_p + q$ cioè

$f(x_p) = f'(x_p)\cdot x_p + q$

$$f(x) = e^{\frac{ax-b}{x+c}}$$

$$f'(x) = e^{\frac{ax-b}{x+c}} \cdot \frac{a(\cancel{x}+c) - 1(\cancel{ax}-b)}{(x+c)^2} = e^{\frac{ax-b}{x+c}} \cdot \boxed{\frac{ac+b}{(x+c)^2}}$$

$(ac+b)(x+c)^{-2}$

$$f''(x) = \underbrace{e^{\frac{ax-b}{x+c}} \cdot \frac{ac+b}{(x+c)^2}} \cdot \underbrace{\frac{ac+b}{(x+c)^2}} + \underbrace{e^{\frac{ax-b}{x+c}}} \cdot \underbrace{(ac+b)\cdot(-2)(x+c)^{-3}}$$

$$\triangleright\ f''(0) = e^{-\frac{b}{c}} \cdot \frac{(ac+b)^2}{c^4} - 2e^{-\frac{b}{c}} \cdot \frac{(ac+b)}{c^3} = 0$$

$$\boxed{\frac{\cancel{e^{-\frac{b}{c}}} \cdot (ac+b)}{c^3} \left(\frac{ac+b}{c} - 2 \right) = 0}$$

$$\triangleright\ f'(0) = \frac{2}{e} \rightsquigarrow \boxed{e^{-\frac{b}{c}} \frac{(ac+b)}{c^2} = \frac{2}{e}}$$

$$\triangleright\ f(0) = f'(0) \cdot 0 + \frac{1}{e} \rightsquigarrow e^{\frac{-b}{c}} = \frac{1}{e} \rightsquigarrow \boxed{b = c}$$

$$\begin{cases} \frac{ac+c}{c^3} \left(\frac{ac+c}{c} - 2 \right) = 0 \\ \frac{ac+c}{c^2} = 2 \end{cases} \rightsquigarrow \begin{cases} \frac{a+1}{c^2} (a-1) = 0 \\ \frac{a+1}{c} = 2 \rightsquigarrow a = 2c-1 \end{cases}$$

$$\frac{2c}{c^2} (2c-1-1) = 0$$

$$\frac{2}{c} \cdot 2(c-1) = 0 \rightsquigarrow \boxed{c=1} \rightarrow \boxed{b=1}, \ \boxed{a=1}$$

Compito del 14

Problema:
- Parametri lim+derivate
- Studio di fz
- Prob max min
- BOH ... Robe a caso
- Boh 2

] 20 punti del Rec

Quesiti:
- 2 lim + 2 derivate
- Rette tangent. a curve
- Comb
- Trigon
- (4) Robe di quinte sup

] 20 pti del Rec

Pag 1964 e seguenti

249 $\int x(3x^2-1)^2\,dx =$

Sost. $3x^2-1 = t$

$x^2 = \frac{t+1}{3} = \frac{t}{3}+\frac{1}{3}$

$2x\,dx = \frac{1}{3}dt$

$x\,dx = \frac{1}{6}dt$

$= \int \frac{1}{6}t^2\,dt = \frac{1}{6}\int t^2 dt$

$= \frac{1}{6}\frac{t^3}{3} + c = \frac{(3x^2-1)^3}{18} + c$

276 $\int \frac{1-x^2}{1+x^2}\,dx = \underbrace{\int \frac{1}{1+x^2}\,dx}_{\boxed{=\operatorname{arctg} x + c}} - \boxed{\int \frac{x^2}{1+x^2}\,dx}$

$\int \frac{x^2}{1+x^2}\,dx \boxed{=} \int \frac{1+x^2-1}{1+x^2}\,dx = \int \frac{1+x^2}{1+x^2}\,dx - \int \frac{1}{1+x^2}\,dx$

Aggiungo e tolgo 1

$$= \int 1\,dx - \int \frac{1}{1+x^2}\,dx = \boxed{x - \operatorname{arctg} x + c}$$

$$\rightsquigarrow \boxed{2\operatorname{arctg} x - x + c}$$

$$x^{-2} \rightsquigarrow -2x^{-3} = -\frac{2}{x^3}$$

298 $$\int \frac{e^{\frac{1}{x^2}}}{x^3}\,dx =$$

Sost. $e^{\frac{1}{x^2}} = t$

$$e^{\frac{1}{x^2}} \cdot \left(-\frac{2}{x^3}\right) dx = dt$$

$$\frac{e^{1/x^2}}{x^3}\,dx = -\frac{dt}{2}$$

$$\int -\frac{dt}{2} = -\frac{1}{2}\int dt =$$

$$= -\frac{1}{2}t + c = -\frac{1}{2}e^{\frac{1}{x^2}} + c$$

373 $$\int \frac{\sin x}{\sqrt{1-\cos x}}\,dx =$$

$\sqrt{1-\cos x} = t$

$$\frac{1}{2}(1-\cos x)^{-\frac{1}{2}} \cdot \sin x\,dx = dt$$

$$\frac{\sin x}{\sqrt{1-\cos x}}\,dx = 2\,dt$$

$$\int 2\,dt = 2t + c =$$

$$\boxed{2\sqrt{1-\cos x} + c}$$

381: $$\int \tan^3 x\,dx = \int \frac{\sin^3 x}{\cos^3 x}\,dx =$$

$\cos x = t$ e $\sin x$ compariva nel differenziale

$$-\sin x\,dx = dt$$

$$\int \frac{\sin x}{\cos^3 x}(\sin^2 x)\,dx =$$

$$\int \frac{\sin x}{\cos^3 x}(1-\cos^2 x)\,dx = \int -\frac{1-t^2}{t^3}\,dt =$$

$$\int\left(-\frac{1}{t^3} + \frac{1}{t}\right)dt = -\frac{t^{-2}}{-2} + \ln|t| + c =$$

t^{-3}

$$\boxed{\frac{1}{2\cos^2 x} + \ln|\cos x| + c}$$

382: $\displaystyle\int \frac{\sin x}{4+\cos^2 x}\,dx =$

Sost: $\cos x = t$

$-\sin x\,dx = dt$

$$\int -\frac{1}{\underbrace{4+t^2}_{\text{Valio for compenire 1}}}\,dt = -\int \frac{1}{4\left(1+\frac{t^2}{4}\right)}\,dt = -\frac{1}{4}\int \frac{1}{1+\frac{t^2}{4}}\,dt$$

Sost: $\frac{t}{2} = z$

$\frac{1}{2}dt = dz$

$dt = 2dz$

$$= -\frac{1}{4}\int \frac{1}{1+z^2}\cdot 2\,dz =$$

$$= -\frac{1}{2}\operatorname{arctg} z + c = -\frac{1}{2}\operatorname{arctg}\frac{t}{2} + c =$$

$$\boxed{-\frac{1}{2}\operatorname{arctg}\frac{\cos x}{2} + c}$$

391: $\displaystyle\int \left(\cos x + \frac{4}{3\sin x}\right)dx =$

$$\underbrace{\int \cos x\,dx}_{\sin x} + \frac{4}{3}\boxed{\int \frac{1}{\sin x}\,dx}$$

$$\int \frac{1}{\sin x}\,dx =$$

Sost: $\operatorname{tg}\frac{x}{2} = t \rightsquigarrow \frac{x}{2} = \operatorname{arctg} t$

$$\boxed{dx = \frac{2}{1+t^2}\,dt}$$

$$\int \frac{\cancel{1+t^2}}{2t}\,\frac{\cancel{2}}{\cancel{1+t^2}}\,dt = \int \frac{1}{t}\,dt = \ln|t| + c$$

$\sin x = \frac{2t}{1+t^2}$ $\quad \cos x = \frac{1-t^2}{1+t^2}$

$$\rightsquigarrow \boxed{\sin x + \ln\left|\operatorname{tg}\frac{x}{2}\right| + c}$$

Integrazione per parti

$$(f \cdot g)' = f' \cdot g + f \cdot g'$$ La integro

$$f \cdot g = \int f' g + \int f g'$$

$$\boxed{\int f' g = f g - \int f g'}$$ ⇝ Int per Parti

Identifico due fz in modo che derivandole e integrandole l'integrale da risolvere diventa più trattabile

407: $\int \underbrace{2x}\,\underbrace{\ln x}\, dx =$

$$x^2 \ln x - \int x^2 \frac{1}{x}\, dx =$$

$$x^2 \ln x - \int x\, dx =$$

$$\boxed{x^2 \ln x - \frac{x^2}{2} + c}$$

$f = x^2 \qquad f' = 2x$

$g = \ln x \qquad g' = \frac{1}{x}$

Es istruttivo:

Per parti

$$\int \ln(x)\, dx = \int 1 \cdot \ln x\, dx \boxed{=}$$

$f = x \qquad f' = 1$

$g = \ln(x) \qquad g' = \frac{1}{x}$

$$x \ln x - \int x \cdot \frac{1}{x}\, dx =$$

$$x \ln x - \int 1\, dx = \boxed{x \ln x - x + c}$$